

Written HW3

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I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

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1. Suppose the number of times a person contracts a cold in a given year is a Poisson with $\lambda = 5$. Also Suppose a new wonder drug has just been marketed that reduces the Poisson parameter to $\lambda = 3$ for 75% of the population; for the other 25%, the drug has no effect. If an individual tries the drug for a year, and gets 2 colds during that time, what is the probability that the drug is beneficial for this individual?

Answer:

To find the probability that the drug is beneficial for this individual we may use Bayes' Theorem to find the likelihood that they are in the group of individuals where the poisson distribution is given by $\lambda = 3$.

$$P(\lambda = 3|\text{drug}) = .75, P(\lambda = 5|\text{drug}) = .25$$

$$P(\lambda = 3|k = 2) = \frac{P(\lambda=3)P(k=2|\lambda=3)}{P(k=2)}$$

$$P(k = 2) = P(\lambda = 3)P(k = 2|\lambda = 3) + P(\lambda = 5)P(k = 2|\lambda = 5)$$

$$= .75 \left(e^{-3} \frac{3^2}{2!} \right) + .25 \left(e^{-5} \frac{5^2}{2!} \right)$$

$$\approx .75 \times .224042 + .25 \times .0842243$$

$$\approx .189088$$

$$P(\lambda = 3|k = 2) \approx \frac{.75 \times .224042}{.189088}$$

$$\approx \boxed{.888641}$$

2. An Interviewer is given a list of people she can interview. She will conduct at most 5 interviews, and each person independently agrees to be interviewed with probability $\frac{2}{3}$.

- (a) If the list has 5 people on it, what's the probability that she gets five interviews?

Answer:

This question follows the format of a Binomial Distribution Probability for the simplest case where $k = n$

$$X \sim \text{Binomial}(5, \frac{2}{3})$$

$$P(X = 5) = \binom{5}{5} \times \left(\frac{2}{3}\right)^5 \times \left(\frac{1}{3}\right)^0$$

$$= \boxed{\frac{32}{243}}$$

- (b) What if the list has 8 people on it?

Answer:

The question remains to be a binomial, though now with $0 < k < n$, resulting in

$$X \sim \text{Binomial}(8, \frac{2}{3})$$

$$P(X = 5) = \binom{8}{5} \times \left(\frac{2}{3}\right)^5 \times \left(\frac{1}{3}\right)^3$$

$$= \boxed{\frac{1792}{6561}}$$

- (c) If the list has 8 people on it, and she begins at the top, what is the probability that she interviews the 7th person?

Answer:

In this case there are 6 possible people preceding the 7th interviewee
 $X \sim \text{Binomial}(6, \frac{2}{3})$

$$P(\text{7th person interviewed}) = \frac{2}{3} \times \sum_{i=0}^4 P(X = i)$$

$$P(X = i) = \binom{6}{i} \times \left(\frac{2}{3}\right)^i \times \left(\frac{1}{3}\right)^{6-i}$$

$$\sum_{i=0}^4 P(X = i) =$$

$$\binom{6}{0} \times \left(\frac{2}{3}\right)^0 \times \left(\frac{1}{3}\right)^6$$

$$+ \binom{6}{1} \times \left(\frac{2}{3}\right)^1 \times \left(\frac{1}{3}\right)^5$$

$$+ \binom{6}{2} \times \left(\frac{2}{3}\right)^2 \times \left(\frac{1}{3}\right)^4$$

$$+ \binom{6}{3} \times \left(\frac{2}{3}\right)^3 \times \left(\frac{1}{3}\right)^3$$

$$+ \binom{6}{4} \times \left(\frac{2}{3}\right)^4 \times \left(\frac{1}{3}\right)^2$$

$$= \frac{1}{3^6} + \frac{6 \times 2}{3^6} + \frac{15 \times 2^2}{3^6} + \frac{20 \times 2^3}{3^6} + \frac{15 \times 2^4}{3^6}$$

$$= \frac{473}{729}$$

$$P(\text{7th person interviewed}) = \frac{2}{3} \times \frac{473}{729} = \boxed{\frac{946}{2187}} \approx 0.432556$$

To verify, I ran the following Python script, which returned 0.433336

```

1 import random
2
3 count = 0
4 trials = 1000000
5
6 for _ in range(trials):
7     interviews = 0
8
9     # Go through persons 1-6
10    for person in range(1, 7):
11        if random.random() < 2/3: # Person says yes
12            interviews += 1
13            if interviews == 5: # Hit the limit
14                break
15
16    # Check if we can reach person 7
17    if interviews < 5:
18        # Ask person 7
19        if random.random() < 2/3: # Person 7 says yes
20            count += 1
21
22 probability = count / trials
23 print(f"P(interviews person 7) = {probability}")
24

```

(d) As in part (c), what is the probability she interviews the 8th person?

Answer:

Like in (c), we are looking for the probability where the j^{th} positioned potential interviewee is interviewed, where now $j = 8$. We can then substitute in $j - 1 = 7$ for the number of preceding potential interviewees and recalculate to find:

$$X \sim \text{Binomial}(7, \frac{2}{3})$$

$$P(\text{8th person interviewed}) = \frac{2}{3} \times \sum_{i=0}^7 P(X = i)$$

$$P(X = i) = \binom{7}{i} \times \left(\frac{2}{3}\right)^i \times \left(\frac{1}{3}\right)^{7-i}$$

$$\Rightarrow P(\text{8th person interviewed}) = \frac{1878}{6561} \approx 0.28624$$

Then, modifying the previous Python script to be `range(1,8)` on line 10 and recomputing, the approximation verifies with a result of 0.285745